THE CLASS $\mathcal{G}$ BEYOND DEGREE FIVE: EXACT WITNESSES, THE 2-UNRAMIFIED CLASSIFICATION IN DEGREES 6–8, AND THE BOTTOM OF THE VARIANCE SPECTRUM

ABSTRACT. Let $\mathcal{G}$ be the class of totally real fields K with $2\mathcal{O}_K^{\succ 0} \subseteq \Sigma\mathcal{O}_K^2$. Its members of degree at most 5 are known (eight fields), and it is expected that there are no others. This note records a computational and theoretical attack on degrees 6, 7, 8 and on the structure of small-variance fields.

(i) We give an exact decision procedure for "$2\alpha \in \Sigma\mathcal{O}_K^2$" and a search for *exceptional* elements (witnesses), and run it on every totally real field of degree 3–8 in the LMFDB lists (193,467 cubic, 13,073 quartic, 9,494 quintic, 9,815 sextic, 301 septic and 920 octic fields). In degrees ≤ 5 it reproduces the Kala–Yatsyna classification exactly; in degrees 6, 7 and 8 every field has an explicit, independently certified witness.

(ii) We show that for fields in which 2 is unramified, membership in $\mathcal{G}$ implies every necessary condition (budget–cost, sharpened rigidity, square dichotomy, and their relative versions) used in the classification of totally real fields of degree 6–8 with a universal $\mathbb{Z}$-form. Combining this with that classification and with explicit $\mathcal{G}$-witnesses for all the fields it leaves (the ones that it excludes only through condition (N)), we obtain: *no totally real field of degree* 6, 7 *or* 8 *in which* 2 *is unramified lies in* $\mathcal{G}$.

(iii) We determine the bottom of the variance spectrum. All totally real algebraic integers with $\mathrm{Var}(x) \leq 2$ are listed in degrees ≤ 10, and an auxiliary-function (linear programming) certificate, verified in interval arithmetic, shows that every other totally real algebraic integer has $\mathrm{Var}(x) \geq 1.69$. Consequently the values of Var below 1.69 are exactly 0, $\frac{5}{4}$, $\frac{14}{9}$, $\frac{27}{16}$, and a totally real field has $\nu(K) < 1.69$ if and only if it contains $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\zeta_7)^+$ or the quartic field of discriminant 725. As a by-product, $\liminf_{\deg x\to\infty} \mathrm{Var}(x) \geq 1.69$, which improves the bound $\sqrt{e} = 1.6487\ldots$ quoted in the working notes.

(iv) We record shape and arithmetic-statistics data from the census, explicit witnesses for the residual real cyclotomic fields F_{36}, F_{44}, F_{52} left open by the dyadic criterion, and a reduction of the remaining problem (sextic fields with 2 ramified) to explicitly described cases.

CONTENTS

1. Introduction 2
Results 2
2. Anatomy of a representation 3
2.1. The unramified case, and relative versions 4
3. An exact decision procedure, and certificates 4
3.1. Deciding $\beta \in \Sigma\mathcal{O}_K^2$ 4
3.2. Rigorous signs 4
3.3. The witness search 4
3.4. Certificates 5
4. The census in degrees 3–8 5
5. Fields with 2 unramified in degrees 6, 7, 8 5
6. The bottom of the variance spectrum 6
6.1. Enumeration 6
6.2. Auxiliary functions 6
6.3. Consequences for $\mathcal{G}$ 8

Date: November 5, 2026.

7. Shapes and arithmetic statistics 8
8. The residual real cyclotomic fields 9
9. The sextic problem with 2 ramified 9
10. Remarks on the working notes 9
11. Reproducibility 10
References 10

1. Introduction

For a totally real field K of degree d write $\mathcal{O}_K^{\succ 0}$ for its totally positive integers and $\Sigma\mathcal{O}_K^2$ for the sums of squares. Put

$$\mathcal{G} = \{K \text{ totally real}: \ 2\mathcal{O}_K^{\succ 0} \subseteq \Sigma\mathcal{O}_K^2\}.$$

Kala and Yatsyna [4] proved that the members of degree at most 5 are

$$\mathbb{Q},\ \mathbb{Q}(\sqrt{2}),\ \mathbb{Q}(\sqrt{3}),\ \mathbb{Q}(\sqrt{5}),\ \mathbb{Q}(\sqrt{2},\sqrt{5}),\ \mathbb{Q}(\zeta_{20})^+,\ \mathbb{Q}(\zeta_7)^+,\ \mathbb{Q}(\rho) \quad (\rho^3 = 4\rho + 2), \tag{1}$$

and it is expected that (1) is all of $\mathcal{G}$. The working notes [9, 10, 11] develop the *trace variance* $\mathrm{Var}(x) = \tau(x^2) - \tau(x)^2$, $\tau = \frac{1}{d}\mathrm{Tr}_{K/\mathbb{Q}}$, as the natural height for this problem, and the companion repository [13] pushes the related *lifting problem* (universal $\mathbb{Z}$-forms) through degrees 6, 7 and 8 using the same invariants. The purpose of this note is to transfer as much as possible of that machinery to $\mathcal{G}$, to test it against complete field tables, and to understand the fields of small variance, which are the only places where members of $\mathcal{G}$ can live.

Throughout, $\Lambda_K = \mathcal{O}_K/\mathbb{Z}$ with the positive definite form Var is the *variance lattice* (up to scaling it is the shape of K), and

$$\nu(K) = \min_{x \in \mathcal{O}_K \setminus \mathbb{Z}} \mathrm{Var}(x), \qquad \nu_2(K) = \min_{x \in (\mathbb{Z}+\mathfrak{r}) \setminus \mathbb{Z}} \mathrm{Var}(x), \qquad t(K) = \min_{\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}} \tau(\alpha),$$

where $\mathfrak{r} = \prod_{\mathfrak{p}|2} \mathfrak{p}^{\lceil e_\mathfrak{p}/2 \rceil}$ is the dyadic square-root radical; one has $2\mathcal{O}_K \subseteq \mathfrak{r}$, $\nu \le \nu_2 \le 4\nu$, and $\nu_2 = 4\nu$ when 2 is unramified.

Results.

(1) *Anatomy of a representation* (§2). A complete description of what a representation $2\alpha = \sum x_i^2$ looks like in the low-trace range, valid without any hypothesis on 2. It contains the budget–cost inequality and rigidity of [9] and adds a *half-square dichotomy*: if $\tau(\alpha) < \nu(K)$ and $K \in \mathcal{G}$ then $2\alpha - c$ is a square in $\mathcal{O}_K$ for an integer $0 \le c < 2\min_j \sigma_j(\alpha)$.
(2) *An exact witness search* (§3) and a census of all LMFDB fields of degree 3–8 (§4); Theorem 4.1.
(3) *The 2-unramified classification in degrees* 6–8 (§5); Theorem 5.2.
(4) *The bottom of the variance spectrum* (§6); Theorems 6.1 and 6.2.
(5) *Shapes and statistics* (§7), *residual cyclotomic fields* (§8), the *ramified sextic problem* (§9), and corrections to the working notes (§10).

All computations use PARI/GP 2.15.4 [8]; the code and all outputs are in the repository accompanying this note (§11). Every exclusion of a field rests on an explicit element whose exceptional status is certified by exact arithmetic, independently of the (GRH-conditional) class group and unit computations used to *find* it.

2. Anatomy of a representation

Lemma 2.1. *Let K be totally real, $\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}$, and $2\alpha = \sum_{i=1}^{N} x_i^2$ with $x_i \in \mathcal{O}_K \setminus \{0\}$. Let $S = \{i : x_i \notin \mathbb{Z}\}$. Then $S \neq \emptyset$, $\sum_i x_i \in \mathfrak{r}$, and*

$$2\tau(\alpha) = \sum_{i \in S} \big(\mathrm{Var}(x_i) + \tau(x_i)^2\big) + \sum_{i \notin S} x_i^2. \tag{2}$$

Moreover:

1. *If $|S| = 1$, say $S = \{1\}$, put $c = \sum_{i \geq 2} x_i^2 \in \mathbb{Z}_{\geq 0}$. Then $2\alpha - c = x_1^2$, $c < 2\min_j \sigma_j(\alpha)$, $x_1 \in (c + \mathfrak{r}) \setminus \mathbb{Z}$, and $2\tau(\alpha) \geq \nu_2(K) + c$.*
2. *If $|S| = 2$, say $S = \{1, 2\}$, and $\tau(\alpha) < \frac{1}{2}\nu_2(K)$, then $x_2 = \varepsilon x_1 + k$ with $\varepsilon = \pm 1$, $k \in \mathbb{Z}$, and*

$$4\alpha = (2x_1 + \varepsilon k)^2 + \kappa, \qquad \kappa = k^2 + 2\sum_{i \geq 3} x_i^2 \in \mathbb{Z}_{\geq 0}, \quad \kappa \equiv 0, 3 \pmod 4,$$

   with $\tau(\alpha) \geq \mathrm{Var}(x_1) + \kappa/4$. If $\kappa = 0$ then $\alpha = x_1^2$.
3. *If $|S| \geq 3$ then $\tau(\alpha) \geq \frac{3}{2}\nu(K)$.*

Proof. If $S = \emptyset$ then $2\alpha \in \mathbb{Z}$, so $\alpha \in \mathbb{Z}$. The coupling $\sum_i x_i \in \mathfrak{r}$ is [9, dyadic coupling lemma]: $s = \sum x_i$ has $s^2 \equiv \sum x_i^2 \equiv 0 \pmod{2\mathcal{O}_K}$, so $v_{\mathfrak{p}}(s) \geq \lceil e_{\mathfrak{p}}/2 \rceil$ for every $\mathfrak{p} \mid 2$. Identity (2) is τ of the representation, with $\tau(x^2) = \mathrm{Var}(x) + \tau(x)^2$.

(1) $x_1^2 = 2\alpha - c$ is totally positive, so $c < 2\sigma_j(\alpha)$ for all j. For integers $n \equiv n^2 \pmod 2$, hence $\sum_{i \geq 2} x_i \equiv c \pmod 2$; as $2 \in \mathfrak{r}$, the coupling gives $x_1 \equiv -\sum_{i \geq 2} x_i \equiv c \pmod{\mathfrak{r}}$. Thus $x_1 \in (\mathbb{Z} + \mathfrak{r}) \setminus \mathbb{Z}$, so $\mathrm{Var}(x_1) \geq \nu_2$, and (2) gives $2\tau(\alpha) \geq \mathrm{Var}(x_1) + c$.

(2) Put $c_0 = \sum_{i \geq 3} x_i \in \mathbb{Z}$. The coupling gives $x_1 + x_2 \in -c_0 + \mathfrak{r}$, and $2x_2 \in \mathfrak{r}$ gives $x_1 - x_2 \in -c_0 + \mathfrak{r}$. If neither $x_1 + x_2$ nor $x_1 - x_2$ is rational, both have variance $\geq \nu_2$, and the parallelogram law for the quadratic form Var gives $2\nu_2 \leq \mathrm{Var}(x_1+x_2)+\mathrm{Var}(x_1-x_2) = 2\,\mathrm{Var}(x_1)+2\,\mathrm{Var}(x_2) \leq 2\tau(x_1^2)+2\tau(x_2^2) \leq 4\tau(\alpha)$, contrary to the hypothesis. Hence $x_2 = \varepsilon x_1 + k$. Expanding, $2\alpha = 2x_1^2 + 2\varepsilon k x_1 + k^2 + M$ with $M = \sum_{i \geq 3} x_i^2$, so $4\alpha = (2x_1 + \varepsilon k)^2 + k^2 + 2M$. Since $k^2 + M = 2(\alpha - x_1^2 - \varepsilon k x_1) \in 2\mathcal{O}_K \cap \mathbb{Z}$, $M \equiv k^2 \pmod 2$ and $\kappa \equiv 3k^2 \pmod 4$. Finally $\tau(\alpha) = \tau\big((x_1 + \varepsilon k/2)^2\big) + \kappa/4 \geq \mathrm{Var}(x_1) + \kappa/4$, and $\kappa = 0$ forces $k = 0 = M$, i.e. $\alpha = x_1^2$.

(3) Three summands of (2) are each $\geq \nu$. □

Corollary 2.2 ($\mathcal{G}$-dichotomies)**.** *Let $K \in \mathcal{G}$, $K \neq \mathbb{Q}$, and $\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}$. Write $\nu = \nu(K)$, $\nu_2 = \nu_2(K)$.*

1. (budget–cost [9, budget–cost inequality]) $\tau(\alpha) \geq \min(\nu, \frac{1}{2}\nu_2)$.
2. (half-square dichotomy) *If $\tau(\alpha) < \nu$, then $2\alpha - c = y^2$ for some $c \in \mathbb{Z}$ with $0 \leq c < 2\min_j \sigma_j(\alpha)$ and some $y \in (c + \mathfrak{r}) \setminus \mathbb{Z}$; moreover $2\tau(\alpha) \geq \nu_2 + c$.*
3. (square window) *If $\tau(\alpha) < \min(\frac{3}{2}\nu,\ \nu + \frac{3}{4},\ \frac{1}{2}\nu_2)$, then α is a square in $\mathcal{O}_K$.*
4. (witness by trace) *If $\tau(\alpha) < \min(\nu, \frac{1}{2}\nu_2)$ for some $\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}$ of an arbitrary totally real K, then $2\alpha \notin \Sigma\mathcal{O}_K^2$; in particular $K \notin \mathcal{G}$, and α itself is a witness.*

Proof. (1), (2), (4) are immediate from Lemma 2.1. For (3): $|S| \geq 3$ is excluded by $\tau < \frac{3}{2}\nu$, and $|S| = 1$ by $\tau < \frac{1}{2}\nu_2$ (Lemma 2.1(1)); so $|S| = 2$, and $\nu + \kappa/4 \leq \tau(\alpha) < \nu + \frac{3}{4}$ forces $\kappa \in \{0, 1, 2\}$, hence $\kappa = 0$ by the congruence, hence $\alpha = x_1^2$. □

Remark 2.3. Apply (2) to the *lower rung* $\alpha_- = x - n$ of any $x \in \mathcal{O}_K \setminus \mathbb{Z}$, where $n = \lceil \theta_1 \rceil - 1$ and θ_1 is the least conjugate of x. Then $\min_j \sigma_j(\alpha_-) = \theta_1 - n \in (0, 1]$, so $c \in \{0, 1\}$: *if $\tau(x - n) < \nu$ then $2x - 2n - c$ is a square in $\mathcal{O}_K$ for some $c \in \{0, 1\}$.* This is the $\mathcal{G}$-analogue of the "rung split" of [13, degree-8 rung split], and it is the only way in which the lower rung can be cheap. The same holds for the upper rung $\lfloor \theta_d \rfloor + 1 - x$.

2.1. **The unramified case, and relative versions.** If 2 is unramified then $\mathfrak{r} = 2\mathcal{O}_K$, $(\mathbb{Z} + \mathfrak{r})/\mathbb{Z} = 2\Lambda_K$, and $\nu_2 = 4\nu$. Then Corollary 2.2(1) reads $\tau(\alpha) \geq \nu$, the half-square alternative never occurs (it forces $\tau(\alpha) \geq 2\nu$), and the window of (3) is $\min(\frac{3}{2}\nu, \nu + \frac{3}{4})$.

Let $F \subsetneq K$ be a subfield, $\mathrm{Var}_{K/F}$ the relative variance (an F-valued totally nonnegative quadratic form vanishing exactly on F), and $\nu_{K/F} = \min\{\tau_F(\mathrm{Var}_{K/F}\, x) : x \in \mathcal{O}_K \setminus \mathcal{O}_F\}$; by the law of total variance [9, ANOVA identity], $\tau(x^2) \geq \tau_F(\mathrm{Var}_{K/F}\, x)$.

Proposition 2.4 (relative version, 2 unramified)**.** *Let $K \in \mathcal{G}$ with 2 unramified in K, $F \subsetneq K$, and $\alpha \in \mathcal{O}_K^{\succ 0} \setminus F$. Then $\tau(\alpha) \geq \nu_{K/F}$. If moreover $\tau(\alpha) < \frac{3}{2}\nu_{K/F}$, then $\alpha = y^2 + \gamma y + \beta$ with $y \in \mathcal{O}_K \setminus \mathcal{O}_F$, $\beta, \gamma \in \mathcal{O}_F$, $\tau_F(\mathrm{Var}_{K/F}\, y) \leq \tau(\alpha)$, and $4\beta - \gamma^2$ totally nonnegative.*

Proof. Let $S_F = \{i : x_i \notin F\} \neq \emptyset$. If $|S_F| \geq 2$ then $\tau(\alpha) \geq \nu_{K/F}$; if $|S_F| \geq 3$ then $\tau(\alpha) \geq \frac{3}{2}\nu_{K/F}$. If $S_F = \{1\}$, the coupling (with $\mathfrak{r} = 2\mathcal{O}_K$) gives $x_1 \in \mathcal{O}_F + 2\mathcal{O}_K$, so $x_1 = f + 2w$ with $w \notin F$ and $\tau(x_1^2) \geq 4\nu_{K/F}$, i.e. $\tau(\alpha) \geq 2\nu_{K/F}$. If $S_F = \{1, 2\}$, then $x_1 \pm x_2 \in \mathcal{O}_F + 2\mathcal{O}_K$, and if both are outside F the parallelogram law for $\tau_F \circ \mathrm{Var}_{K/F}$ gives $\tau(\alpha) \geq 2\nu_{K/F}$. Otherwise $x_2 = \pm x_1 + \gamma$ with $\gamma \in \mathcal{O}_F$; then $\alpha = x_1^2 \pm \gamma x_1 + \beta$ with $2\beta = \gamma^2 + \sum_{i \geq 3} x_i^2$, $\beta \in \mathcal{O}_K \cap F = \mathcal{O}_F$, and $4\beta - \gamma^2 = \gamma^2 + 2\sum_{i \geq 3} x_i^2 \succeq 0$. □

3. An exact decision procedure, and certificates

3.1. **Deciding $\beta \in \Sigma\mathcal{O}_K^2$.** Let $\beta \in \mathcal{O}_K^{\succ 0}$. Every summand x of a representation $\beta = \sum x_i^2$ satisfies $\sigma_j(x)^2 \leq \sigma_j(\beta)$ for all j, hence lies in the finite set

$$X(\beta) = \Big\{x \in \mathcal{O}_K \setminus \{0\} :\ \textstyle\sum_j \sigma_j(x)^2/\sigma_j(\beta) \leq d\Big\} \cap \{x : \beta - x^2 \succeq 0\}.$$

We enumerate the ellipsoid by Fincke–Pohst (PARI's `qfminim` on the weighted Gram matrix, LLL-reduced, at a working precision adapted to the spread of the conjugates of β) and test $\beta - x^2 \succeq 0$ exactly. A depth-first search over multisets, with a memo on remainders and a direct look-up for one and two squares, decides whether $\beta \in \Sigma\mathcal{O}_K^2$. Every representation found is re-verified by exact multiplication.

3.2. **Rigorous signs.** All decisions about signs of conjugates are exact. The embedding matrix is computed to ≥ 100 digits, and a numerical sign is accepted only when the computed value exceeds an explicit bound $(\|x\|_1 + 1)\max|M_{ij}|\, 10^{-90}$ for the error; otherwise the sign is decided from a certified high-precision evaluation (precision doubled until $|v|$ exceeds a rigorous evaluation error) or by a Sturm sequence on the characteristic polynomial. This matters: in large-regulator fields the totally positive generators of small ideals are unavoidably unbalanced (conjugates of size $10^{\pm 50}$ occur for cubic fields of discriminant $\approx 2 \cdot 10^6$), and a first version of the code that used a fixed tolerance produced wrong sign vectors of fundamental units there. The bug was caught by the final certification step below.

3.3. **The witness search.** For a field K we try, in order: (U) the *unit criterion*: every $u \in U_K^+$ with $u, 2u \notin K^{\times 2}$ is a witness, since $\mathrm{N}(u) = 1$ allows at most two squares and equality in O'Meara's superadditivity forces u or $2u$ to be a square [9, unit dichotomy lemma]; (T) all $\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}$ with $\tau(\alpha) \leq T$, enumerated in Λ_K using $\mathrm{Var}(\alpha) \leq (d-1)\tau(\alpha)^2$ and the least totally positive shift of each class (Corollary 2.2(4) explains why such elements are natural witnesses); (N) for every ideal of norm $\leq N_{\max}$ that is narrowly principal, every totally positive generator modulo U_K^2, reduced in trace by squares of units.

TABLE 1. The census. U, T, N: number of fields whose first witness is a unit, a small-trace element, or a small-norm generator. "max N" is the largest norm of a first witness of type N.

d	fields	max d_K	U	T	N	left	$h^+ = h$	max N
4	13073	$1.00 \cdot 10^6$	7113	1118	4840	2 (1600,2000)	30.7%	81
5	9494	$1.00 \cdot 10^7$	5170	3097	1227	0	45.5%	17
6	9815	$1.00 \cdot 10^8$	5024	4413	378	0	40.6%	43
7	301	$2.14 \cdot 10^8$	156	145	0	0	48.2%	0

3.4. **Certificates.** A witness α is accepted only after (a) total positivity is re-proved by a Sturm sequence of its characteristic polynomial, with no floating point, and (b) the decision procedure is rerun from scratch on 2α and finds no representation. Neither step uses the class group or the units, so every exclusion below is unconditional, although the search that found the witness used `bnfinit` (correct under GRH).

4. The census in degrees 3–8

We ran the search on the LMFDB lists of totally real fields [5] as mirrored in [14] (the lists used by the `sails` project). The bounds were $T = 3$ ($T = 2$ in degree 3) and $N_{\max} = 64, 128, 300, 300, 400$ in degrees $4, 5, 6, 7, 8$ ($N_{\max} = 64$ in degree 3); the few survivors were rerun with $N_{\max} = 1500$.

Theorem 4.1. *Every one of the 9815 fields of degree 6 ($d_K \le 1.00 \cdot 10^8$), 301 fields of degree 7 ($d_K \le 2.14 \cdot 10^8$) in the LMFDB lists has an explicit element $\alpha \in \mathcal{O}_K^{\succ 0}$ with $2\alpha \notin \Sigma\mathcal{O}_K^2$, certified as in §3; none of these fields lies in $\mathcal{G}$. In degrees 4, 5 the fields without a witness are exactly the members of (1) in the lists.*

Remark 4.2 (validation against [4])**.** In degrees 3, 4, 5 the procedure finds a witness for every field except exactly the members in (1) (discriminants $49, 148$; $1600, 2000$; none), which survive up to $N_{\max} = 1500$. Since our implementation shares no code with [4], this is an independent check of their classification on these lists, and conversely their theorem is a check of our code: a single false witness or a missed representation in a member would have shown up here.

Remark 4.3 (the extreme cases)**.** The fields that need the largest witnesses are imprimitive fields with a subfield of very small variance and a large discriminant: the seven quartic fields with $d_K \in [5.7 \cdot 10^5, 10^6]$ that survive $N_{\max} = 64$ all contain $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(\sqrt{2})$, have normalised shape minimum ≈ 0.09 (against a median 0.58), and need witnesses of norm 71–81. This is the "inherited witness" phenomenon of [9, "What remains, and why"] seen in data: a small-variance subfield blocks every trace-based witness, and only norm-based witnesses remain, whose size grows with d_K.

5. Fields with 2 unramified in degrees 6, 7, 8

The repository [13] proves that no totally real field of degree 6, 7 or 8 admits a universal $\mathbb{Z}$-form. Its proof uses the following necessary conditions on a field K (we refer to the notes in [13] for the precise statements):

(BC) $\tau(\alpha) \ge \nu(K)$ for all $\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}$, and its integer-part consequences (downspread, the quadratic filter);

(RG) rigidity: $\tau(\alpha) < \frac{3}{2}\nu(K)$ implies $\alpha = y^2 + cy + b$ with $y \notin \mathbb{Z}$, $\mathrm{Var}(y) \le \tau(\alpha)$, $\kappa = 4b - c^2 \ge 0$, $\kappa \equiv 0, 3 \pmod 4$; in particular the square dichotomy $\tau(\alpha) < \min(\frac{3}{2}\nu, \nu + \frac{3}{4}) \Rightarrow \alpha \in \mathcal{O}_K^2$;

TABLE 2. $\mathcal{G}$-witnesses for the fields left by the pipeline of [13] that have 2 unramified, in degree 6, and for the two octic fields excluded there only by (N).

d_K	witness α (minimal polynomial)	$\mathrm{N}_{K/\mathbb{Q}}\,\alpha$
300125	$x^6-14x^5+73x^4-182x^3+227x^2-133x+29$	29
485125	$x^6-13x^5+64x^4-151x^3+177x^2-96x+19$	19
820125	x^3-6x^2+9x-3	9
1416125	$x^6-14x^5+75x^4-191x^3+232x^2-117x+19$	19
3162625	$x^6-15x^5+86x^4-234x^3+299x^2-142x+4$	4
442050625	$x^4-9x^3+27x^2-31x+11$	121
656505625	$x^4-9x^3+27x^2-31x+11$	121

(REL) the relative versions of (BC) and (RG) over every subfield;
and, only at the very end, condition (N) at explicit elements.

Proposition 5.1. *If $K\in\mathcal{G}$ and 2 is unramified in K, then K satisfies (BC), (RG) and (REL).*

Proof. (BC) and (RG) are Corollary 2.2 with $\nu_2=4\nu$ together with Lemma 2.1(2) (which gives exactly the shape y^2+cy+b with $\kappa=k^2+2\sum_{i\ge3}x_i^2\ge0$); (REL) is Proposition 2.4. □

Theorem 5.2. *No totally real field of degree 6, 7 or 8 in which 2 is unramified belongs to $\mathcal{G}$.*

Proof. By Proposition 5.1, such a field passes every test of [13] that precedes the final condition-(N) step. That computation leaves, in degree 6, the 18 fields of [13, degree-6 Case B] (Case B; Case A is empty); in degree 7, nothing; in degree 8, the four fields of Case A and the 390 fields of Case B over $\mathbb{Q}(\sqrt5)$. For each of these $18+4+390$ fields we found an explicit $\alpha\in\mathcal{O}_K^{\succ0}$ with $2\alpha\notin\Sigma\mathcal{O}_K^2$, certified as in §3 (files `results/g2o_d6.txt`, `g2o_d8A.txt`, `g2o_d8B.txt`); Table 2 lists the ones with 2 unramified in degree 6 and the two octic fields that survive all tests of [13] except (N). □

Remark 5.3. (a) The proof uses the computations of [13] as a black box; we have checked that each exclusion test there is one of (BC), (RG), (REL), and reproduced its degree-6 and degree-7 runs (§11). (b) Most of the fields in the list have 2 ramified; they are not covered by Proposition 5.1, but our witnesses exclude them from $\mathcal{G}$ anyway. (c) For $\mathbb{Q}(\sqrt5,\zeta_7^+)$ ($d_K=300125$) and the two octic fields, the $\mathcal{G}$-witness is the same element at which (N) fails in [13]: the elements of trace 14, norm 29 and of trace 18 (in the quartic subfield of discriminant 725), respectively.

6. THE BOTTOM OF THE VARIANCE SPECTRUM

6.1. **Enumeration.** Up to $x\mapsto\pm x+k$ and conjugation, a totally real algebraic integer of degree e is a real-rooted monic $f\in\mathbb{Z}[t]$ with $a_1\in\{0,-1,\dots,-\lfloor e/2\rfloor\}$, and $e\,\mathrm{Var}(x)=\frac{e-1}{e}a_1^2-2a_2$. We enumerated all such f with $\mathrm{Var}\le2$ (Robinson's method, the enumerator of [13] in its unfiltered mode) for $e\le10$ and kept the irreducible ones.

6.2. **Auxiliary functions.** Let $I=[s_0,s_1]\subseteq[0,\frac12]$, $Q_1,\dots,Q_k\in\mathbb{Z}[t]$ and $c_1,\dots,c_k\ge0$ be such that

$$f_I(t)=\mathrm{dist}(t,I)^2-\sum_i c_i\log|Q_i(t)|\ \ge\ m_I\qquad(t\in\mathbb{R}).\tag{3}$$

If x is a totally real algebraic integer with $\tau(x)\in I$ and $Q_i(x)\neq0$ for all i, then averaging (3) over the conjugates of x and using $|\,\mathrm{N}(Q_i(x))|\ge1$ gives $\mathrm{Var}(x)\ge\frac1e\sum_j\mathrm{dist}(\theta_j,I)^2\ge m_I$. Every x can be normalised by $x\mapsto\pm x+k$ to have $\tau(x)\in[0,\frac12]$.

TABLE 3. The variance spectrum below 1.9 in degrees ≤ 10: value, degrees in which it occurs, number of polynomials (up to $x \mapsto \pm x + k$).

Var	decimal	degrees	#
5/4	1.2500	2	1
14/9	1.5556	3	1
27/16	1.6875	4	1
7/4	1.7500	4,8	3
44/25	1.7600	5	1
16/9	1.7778	9	1
9/5	1.8000	10	3
65/36	1.8056	6	2
90/49	1.8367	7	3
46/25	1.8400	5,10	9
37/20	1.8500	10	12
119/64	1.8594	8	8
152/81	1.8765	9	15
17/9	1.8889	6	4
189/100	1.8900	10	43

We took for the Q_i the linear polynomials $t - k$ and all integer shifts and reflections, with roots in $[-4.5, 5]$, of the irreducible polynomials of degree ≤ 7 found in the enumeration (959 polynomials), solved the linear program for the c_i on a grid, rounded the c_i down to rationals, and *verified* (3) on all of $\mathbb{R}$: on $[-7, 7.5]$ by interval arithmetic (`mpmath.iv`, 120 bits) with adaptive subdivision, using $\log|Q_i(t)| \le \log \max_{\text{cell}} |Q_i|$; outside, by the explicit increasing minorant $(|t| - \frac{1}{2})^2 - \sum_i c_i(\log \|Q_i\|_1 + \deg Q_i \log|t|)$.

Theorem 6.1. *Let $m^* = \frac{33839}{20000} = 1.69195$. There is an explicit set $\mathcal{Q}$ of 109 irreducible polynomials in $\mathbb{Z}[t]$, of degrees ≤ 7 (file `results/lp/cert_deg7.json`), such that every totally real algebraic integer x which is not, after $x \mapsto \pm x + k$ ($k \in \mathbb{Z}$), a root of a member of $\mathcal{Q}$ satisfies*

$$\mathrm{Var}(x) \;\ge\; m^*.$$

The members of $\mathcal{Q}$ have degree ≤ 7, so their roots appear in the enumeration, and among them only the following have variance $< m^*$: $\frac{1+\sqrt{5}}{2}$ ($\mathrm{Var} = \frac{5}{4}$), $2\cos\frac{2\pi}{7}$ ($\frac{14}{9}$), and a root of $t^4 - t^3 - 3t^2 + t + 1$ ($\frac{27}{16}$; it generates the totally real quartic field of discriminant 725, the one of least discriminant). Consequently

$$\{\mathrm{Var}(x) : x \text{ a totally real algebraic integer}\} \cap [0, m^*) = \{0, \tfrac{5}{4}, \tfrac{14}{9}, \tfrac{27}{16}\},$$

each value being attained by a unique integer up to $x \mapsto \pm x + k$ and conjugation, and $\mathrm{Var}(x) \ge m^*$ for every totally real algebraic integer of degree ≥ 5. With the 36 polynomials of degree ≤ 5 of `cert_deg5.json` one already gets the bound $\frac{32597}{20000} = 1.62985$.

Theorem 6.2. *Let K be totally real and m^* as in Theorem 6.1. Then $\nu(K) < m^*$ if and only if K contains one of $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\zeta_7)^+$, F_{725} (the quartic field of discriminant 725), and in that case*

$$\nu(K) = \begin{cases} \frac{5}{4} & \text{if } \sqrt{5} \in K, \\ \frac{14}{9} & \text{if } \sqrt{5} \notin K,\ \mathbb{Q}(\zeta_7)^+ \subseteq K, \\ \frac{27}{16} & \text{otherwise.} \end{cases}$$

TABLE 4. Normalised shape minimum $\hat{\nu}$ in the census: quantiles, the Hermite bound γ_{d-1}, medians by type of first witness, and the share of fields with $h^+ = h$ (equivalently $U_K^+ = U_K^2$).

d	$q_{0.01}$	$q_{0.1}$	$q_{0.5}$	$q_{0.9}$	max	γ_{d-1}	U	T	N	$h^+ = h$
4	0.08	0.12	0.58	0.90	1.19	1.26	0.55	0.58	0.62	30.7%
5	0.47	0.55	0.73	0.94	1.20	1.41	0.73	0.72	0.80	45.5%
6	0.27	0.34	0.74	0.95	1.32	1.52	0.74	0.77	0.37	40.6%
7	0.85	0.88	0.97	1.08	1.25	1.67	0.96	0.98	–	48.2%

In particular $\nu(K) \geq m^*$ for every primitive totally real field of degree ≥ 5, and for every totally real field of odd degree not divisible by 3 (no quadratic, cubic or quartic subfield).

Proof. Let $x \in \mathcal{O}_K \setminus \mathbb{Z}$ realise $\nu(K)$. If $\nu(K) < m^*$, Theorem 6.1 and the enumeration identify x, up to $x \mapsto \pm x + k$ and conjugation, as one of the three integers listed, so K contains the corresponding field and $\nu(K) \in \{\frac{5}{4}, \frac{14}{9}, \frac{27}{16}\}$; which value occurs is the least among the subfields contained in K. Conversely each of the three fields contains an element of the stated variance, and ν can only decrease in extensions. □

Remark 6.3. (a) Theorem 6.1 gives $\liminf_{\deg x \to \infty} \mathrm{Var}(x) \geq m^*$; the working notes [9, the limiting-variance theorem] quote $\sqrt{e} = 1.6487\ldots$ (Schur's argument via Hermite discriminants; see also [2, 1]), and the upper bound $\lambda_{\mathrm{SSS}} \leq 1.8984$ comes from Smith's realisation theorem [12, 7]. We have not found a better lower bound in the literature, but have not made an exhaustive search. (b) The degree-≤ 5 certificate already gives $m^* = 1.6298$, enough to show that $\frac{5}{4}$ and $\frac{14}{9}$ are the only nonzero values below 1.6298; this sharpens the "bottom of the spectrum" theorem of [9, bottom of the spectrum], which proves that $(\frac{5}{4}, \frac{14}{9})$ contains no value. (c) The quartic field of discriminant 725 is precisely the subfield on which the two octic survivors of [13] fail (N) and $\mathcal{G}$ (Table 2): small-variance fields are where the hard cases live.

6.3. **Consequences for $\mathcal{G}$.** The eight members (1) have $\nu \in \{\frac{5}{4}, \frac{14}{9}, 2, \frac{20}{9}, 3\}$. By Theorem 6.2, a field $K \in \mathcal{G}$ with $\nu(K) < m^*$ contains $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\zeta_7)^+$ or the quartic field of discriminant 725; every other member has $\nu(K) \geq m^*$, and by the degree-free exclusion [9, degree-free exclusion] its minimal vector has degree e with $m^* \leq \nu(K) \leq 2e + 2\sqrt{e^2 - 1}$.

7. SHAPES AND ARITHMETIC STATISTICS

The lattice $(\Lambda_K, \mathrm{Var})$ is the shape of K up to scaling [6], with determinant d_K/d^d. We record the *normalised shape minimum* $\hat{\nu}(K) = \nu(K)\,(d^d/d_K)^{1/(d-1)} \in (0, \gamma_{d-1}]$.

Table 4 summarises the shapes. Three features stand out. (a) In the prime degrees 5 and 7 the shape minima concentrate in a band around 0.7–1.0, well below the Hermite bound, while in degrees 4, 6, 8 there is a heavy lower tail: these are the imprimitive fields, whose minimal vector lies in a subfield of small variance and whose shape is therefore skewed. (b) In degree 6, witnesses of type T (small trace, i.e. $t(K)$ small compared with ν and ν_2: Corollary 2.2(4)) occur for well-rounded shapes and witnesses of type N for skewed ones: the medians of $\hat{\nu}$ are 0.77 and 0.37. This is the budget–cost mechanism seen statistically: a round shape forces ν to be large compared with the cheapest totally positive element, while a skewed (imprimitive) shape inherits a small ν from a subfield and only a norm-type witness is left. In degree 4 the distinction is not visible. (c) Between one half and two thirds of the fields have $U_K^+ \neq U_K^2$; in odd degree all of these are excluded by the unit criterion, since then (2) is not the square of an ideal and $2u$ can never be a square. The proportion of fields with $h^+ = h$ that we observe (31–48% in degrees 4–7)

is the kind of statistic predicted by the heuristics of Dummit and Voight for signature ranks of units [3]; we have not compared the numbers with their predictions.

8. The residual real cyclotomic fields

The dyadic criterion of [10] leaves, among $m = 4m_0$, the family with $-1 \in \langle 2 \rangle$ modulo m_0 and $\varphi(m_0)/\operatorname{ord}_{m_0}(2)$ odd; for $m_0 \le 99$ these are $m_0 = 3, 5, 9, 11, 13, 19, 25, 27, 29, 37, 43, 53, 59, 61, 67, 81, 83$ (we re-derived this table). The first two are members. Our search gives explicit witnesses for the next ones:

m	m_0	$[F_m : \mathbb{Q}]$	type	$\mathrm{N}\,\alpha$	$\mathrm{Tr}\,\alpha$
36	9	6	trace	24	18
44	11	10	norm	89	28

These exclusions are unconditional (the witnesses are certified as in §3); in particular F_{44} and F_{52} are $q \equiv 3 \pmod 8$ and $q \equiv 5 \pmod 8$, $q < 269$, cases that neither [10] nor the signature law of [9] reaches unconditionally.

9. The sextic problem with 2 ramified

By Theorem 5.2 and the census, a sextic member of $\mathcal{G}$ would have 2 ramified and discriminant beyond the LMFDB list. We record how the minimal-vector method splits in that case. Let x_1 realise ν, with rungs $\alpha_\pm$ as in Remark 2.3.

Regime 1: $\nu_2 \ge 2\nu$. Then Corollary 2.2(1) gives $\tau(\alpha) \ge \nu$ for every $\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}$, so the downspread bound $d\nu \ge 2(\nu-1)^2$ and the quadratic filter hold, and Case A of the enumeration of [13] (with its 15,720 irreducible candidates, $\nu \le 4.79$) covers all such K with $\mathbb{Q}(x_1) = K$. What is lost is rigidity, whose window shrinks to $\tau < \frac{1}{2}\nu_2$ when $\nu_2 < 3\nu$; the 9,785 candidates that [13] excludes by rigidity must instead be tested at the level of the field.

Regime 2: $\nu \le \nu_2 < 2\nu$. Budget–cost only gives $\tau \ge \frac{1}{2}\nu_2$. A cheap rung is then a *half-square*: $2x_1 - 2n - c = y^2$ with $c \in \{0, 1\}$ (Remark 2.3). If one rung is regular, the downspread argument gives $\frac{3}{2}\nu \le \sqrt{2d\nu} + 2$, i.e. $\nu \le 7.77$; if both are half-squares, $y^2 + z^2 = N$ for an integer $N \in [2\nu, 4\nu)$ and $\nu \le 15.75$. In both cases one enumerates y with $p_2(y) = d(2\tau(\alpha_-) - c)$ and $p_4(y) = d\big(4\nu + (2\tau(\alpha_-) - c)^2\big)$ fixed, as in the rung split of [13, degree-8 rung split].

Case B (the minimal vector lies in a quadratic or cubic subfield F). If $[K : F]$ is odd, an element of F that is a square in K is a square in F, so every half-square condition of Corollary 2.2(2) for elements of F must already hold in F. For example $F = \mathbb{Q}(\sqrt{13})$ cannot occur in a sextic: $\alpha = \frac{5+\sqrt{13}}{2}$ has $\tau(\alpha) = \frac{5}{2} < \nu = \frac{13}{4}$ and neither $5 + \sqrt{13}$ nor $4 + \sqrt{13}$ is a square in F.

Problem 9.1. Carry out the regime-2 enumeration and the relative half-square analysis in Case B; together with Theorem 5.2 this would complete the classification of sextic members of $\mathcal{G}$.

10. Remarks on the working notes

We checked several statements of [9, 10] against independent computations.

(1) The values ν, ν_2, t for the members, for $\mathbb{Q}(\sqrt{13})$, and for the cyclic cubic fields of conductor $13, 91, 117, 133$ in [9, shape section] agree with ours.
(2) The table of excluded and surviving $m_0 \le 99$ in [10, table of $m_0 \le 99$] agrees with ours.
(3) The effective Northcott subsection of [9] states that the cubic scan with $d_K \le 3000$ leaves $\{49, 81, 169, 229, 257, 316, 321, 404, 469, 473, 564\}$ "together with the 2-ramified $d_K = 148$". This is the list of cubic fields with $d_K \le 3000$ and $\nu \le 3.7321$ irrespective of ramification;

besides 148, the fields 316, 404, 564 are also 2-ramified and are therefore not covered by the 2-unramified threshold used there.

(4) The limiting-variance theorem of [9] gives $\sqrt{e}$ as the lower bound for $\liminf \mathrm{Var}$; Theorem 6.1 improves it.

11. Reproducibility

The repository contains:

- `src/gtest.gp`: the toolkit (field record, rigorous signs, the decision procedure `sosdecide`, unit criterion, totally positive generators modulo U_K^2, the witness search `gwitness2` with certification, and the invariants ν, ν_2, t);
- `src/census.gp`, `src/runcensus.sh`: the census (one output line per field: discriminant, polynomial, h^+/h, dyadic type, ν, status, witness, norm, time, ν_2, h, regulator, normalised shape minimum, trace of the witness);
- `src/run2o.gp`, `data/2o/`: $\mathcal{G}$-witnesses for the fields left by [13];
- `src/enum6.c` (copied from [13]), `src/varspec.sh`, `src/spectrum.gp`: the variance spectrum; `src/lp/varlp.py`, `src/lp/varlp_cert.py`: the linear programs and their interval-arithmetic verification;
- `src/cyclo.gp`: real cyclotomic fields; `src/regime1.gp`: the ramified sextic Case A test of §9; `src/gen_tables.py`: the tables of this note.

All runs used 4 cores; the full census takes about two hours, dominated by degree 8 and by the 193,467 cubic fields.

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